

The arithmetic mechanics of Collatz excursions: binomial normal form, negative-cycle shadowing, and the composition of record climbs

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Abstract

Let $T(n) = (3n + 1)/2^{e(n)}$, $e(n) = v_2(3n + 1)$, on odd integers. We show that the *climbing phases* of Collatz trajectories are exact algebra rather than statistical accident. (1) *Binomial normal form*: writing $n = u \cdot 2^s - v$ with v odd, the map acts by $T(n) = 3u \cdot 2^{s-e} - v'$ with $v' = (3v - 1)/2^e$, $e = v_2(3v - 1)$ — the subtrahend runs through the $3x-1$ dynamics, i.e. the 2-adic shadow target $-v$ runs through its own (negative) Collatz orbit. Climbs are episodes of this renewal, and the classical Mersenne climb is its unique fixed point $v = 1$. (2) *Exit laws*: for $l = 1$ exits the next run length is $K' = v_2(3^K a + 1) - 1$; lifting-the-exponent gives a one-line *dead-stop theorem* (pure Mersenne climbs never chain) and mod-8 *selection rules* for chaining. These decode the famous orbit of 27 move by move, and explain the classical list $\{27, 31, 41, 47, 55, 63\}$ of all $n < 10^6$ with excursion ratio above 2. (3) *Statistical camouflage*: under the Haar measure on $\mathbb{Z}_2$ the macro-symbol pair (exit valuation, next run length) is *exactly* i.i.d. $\text{Geom}(1/2)^2$, and measured re-entry depths along orbits are exactly geometric — all arithmetic structure is invisible at the symbol level. (4) *Composition of record climbs*: the tilted-measure analysis of the Lagarias–Weiss repeated random walk model yields a closed-form identity $I(4/3) = \log_2 3 - \frac{4}{3}$ for the Kullback–Leibler cost, from which the cheapest sustained climb has halving-exponent mean exactly $4/3$, i.e. $e = 1$ on *three quarters* of climb steps; the longest record climbs measure 0.714 and 0.725. The same identity re-derives the Lagarias–Weiss prediction $t(n) \approx n^2$ for path records and matches the record envelope with fitted slope 0.971.

1 Setup and coding

Throughout, $T(n) = (3n + 1)/2^{e(n)}$ on odd n , $e(n) = v_2(3n + 1) \geq 1$; $\alpha = \log_2 3$. The *itinerary* of n is $(e(T^t n))_{t \geq 0}$, with partial sums $s_m = \sum_{t < m} e(T^t n)$. On the odd 2-adic integers T is continuous and the itinerary determines the point:

Lemma 1.1 (Coding injectivity). [PROVED] *Distinct $x \neq y \in \mathbb{Z}_2^\times$ have itineraries that eventually differ.*

Proof. If the itineraries agree for m steps then $|T^m x - T^m y|_2 = 2^{s_m} |x - y|_2$, which exceeds 1 for large m unless $x = y$, contradicting $T^m x, T^m y \in \mathbb{Z}_2$. $\square$

The itinerary coding of $\mathbb{Z}_2^\times$ (Terras [?], Everett [?]; the conjugacy of Bernstein–Lagarias [?]) transports Haar measure to the law in which the e_t are i.i.d. with $\Pr(e = j) = 2^{-j}$.

*With extensive machine assistance (Claude/Jengo); every numbered claim carries an evidence label, and all computations are reproducible from the public repository github.com/martiendejong/collatz_ai (scripts 177–182).

2 The binomial normal form

Theorem 2.1 (Renewal law). [PROVED + VERIFIED] *Let $n = u \cdot 2^s - v$ with v odd, $0 < v$, and $e := v_2(3v - 1) < s$. Then*

$$T(n) = 3u \cdot 2^{s-e} - v', \quad v' = \frac{3v-1}{2^e}, \quad e(n) = e.$$

Proof. $3n + 1 = 3u \cdot 2^s - (3v - 1)$; since $v_2(3v - 1) = e < s$, the valuation of the sum is e ; divide. (Machine-verified on 10^5 random (u, s, v) ; script 180.) $\square$

Remark 2.2. The subtrahend evolves by $v \mapsto (3v - 1)/2^{v_2(3v-1)}$: the $3x-1$ map, i.e. the ordinary Collatz map on the *negative* integer $-v$. Thus a number within 2-adic distance 2^{-s} of $-v$ *tracks the negative orbit of $-v$* while the power-of-two budget s is spent at rate e per step. The episode ends when the parts collide ($e \geq s$): finite fuel is exact algebra. The case $v = 1$ is the fixed point $v' = (3 - 1)/2 = 1$: the classical folklore climb $2^K a - 1 \rightarrow 3^K a - 1$ through a ones-run.

Corollary 2.3 (Climb rate while shadowing a cycle). [PROVED] *If $-v$ lies on a negative cycle with A odd steps and B halvings (necessarily $B/A < \alpha$, see §??), an episode entered at depth d lasts $\sim d$ halvings and gains $d(A\alpha - B)/B$ bits: 0.585 per halving for the cycle $\{-1\}$, 0.0566 for $\{-5, -7\}$, 0.0082 for the -17 cycle. Since $(A\alpha - B)/B \leq \alpha - 1$ always, no episode repays its own entry depth: the deficit is at least $(2 - \alpha)d = 0.415d$ bits.*

3 Exit laws, dead stop, selection rules

Write an odd n as $n = 2^K a - 1$, $K = v_2(n + 1)$, a odd (the maximal normal-form window). After the K -step run, $3^K a - 1 = 2^l u$ with u odd.

Theorem 3.1 (Exit law for $l = 1$). [PROVED + VERIFIED] *If $l = 1$ then the next run length and multiplier are*

$$K' = v_2(3^K a + 1) - 1, \quad a' = \frac{3^K a + 1}{2^{K'+1}}.$$

Proof. $u = (3^K a - 1)/2$, so $u + 1 = (3^K a + 1)/2$. (Verified on 10^5 cases.) $\square$

Theorem 3.2 (Mersenne dead stop). [PROVED + VERIFIED] *For $a = 1$ and K odd: $l = 1$ and $K' = 1$. A pure Mersenne climb never chains.*

Proof. Lifting the exponent: $v_2(3^K + 1) = v_2(3 + 1) = 2$ for odd K ; apply Theorem ??. (Verified $K \leq 199$.) $\square$

Proposition 3.3 (Mod-8 selection rules). [PROVED + VERIFIED] *A deep restart ($K' \geq 2$) after an $l = 1$ exit occurs iff $3^K a \equiv 7 \pmod{8}$, i.e.*

$$K \text{ even} : a \equiv 7 \pmod{8}, \quad K \text{ odd} : a \equiv 5 \pmod{8}.$$

More generally the channel $c = 3^K a \bmod 8$ determines the exit: $c = 3$: $l = 1$, $K' = 1$ forced; $c = 7$: $l = 1$, $K' \geq 2$; $c = 5$: $l = 2$; $c = 1$: $l \geq 3$. (Asserted at every macro step of the five record orbits below.)

3.1 The orbit of 27, decoded

The rules read the most famous Collatz excursion as deterministic arithmetic (all steps machine-checked; script 177):

$$\begin{aligned}
 &\underbrace{27 = (K=2, a=7)}_{a \equiv 7 \checkmark} \xrightarrow{K'=v_2(64)-1=5} \underbrace{31 = 2^5 - 1}_{\text{Mersenne}} \xrightarrow{\text{dead stop}} 121 \rightarrow \underbrace{91 = (2, 23)}_{a \equiv 7 \checkmark} \xrightarrow{K'=3} 103 = (3, 13) \\
 &\underbrace{(3, 13)}_{a \equiv 5 \checkmark} \xrightarrow{K'=4} 175 = (4, 11) \xrightarrow{a \equiv 3: \text{ rule fails}} \text{dead stop, descent begins.}
 \end{aligned}$$

Three rule-conforming hits, then predicted failure. Moreover the classical list of all $n < 10^6$ with excursion ratio $\rho(n) = \log t(n)/\log n > 2$, namely $\{27, 31, 41, 47, 55, 63\}$ (cf. [?]), consists of the orbit of 27 and its Mersenne neighbours: the outlier list *is* the decoded chain.

3.2 Shadowing measurements

[MEASURED] Along the record orbits 27, 703, 26623, 626331, 837799: the 2-adically closest negative-cycle point advances *in cycle order* during climbs with depth loss exactly e per step (as $|T'|_2 = 2^e$ dictates); mean shadow depth is 5.3–6.8 during climbs vs 4.3–5.1 during descents; over $\sim 8.5 \times 10^4$ pooled steps, shadow depth ≥ 7 flips the mean 3-step growth from -2.1 bits to $+0.3$ bits (Haar reference -1.245). Deep re-entries crystallize as binomials: on the 703 orbit, $32525 = 2^{15} - 3^5$ and $12197 = 3 \cdot 2^{12} - 91$ launch a 13-step climb wave whose subtrahend chain $243 \rightarrow 91 \rightarrow 17$ is the negative orbit of -3^5 , exactly as Theorem ?? prescribes.

4 The sign dichotomy

For any T -cycle with A odd steps, B halvings, the cycle value is $n_1 = C/D$ with $C > 0$ and $D = 2^B - 3^A$, so

$$\text{sign}(n_1) = \text{sign}(2^B - 3^A) :$$

cycles *above* the entropy threshold ($B/A > \alpha$) are positive (zero 2-adic tails), cycles *below* it are negative (ones tails). The negative cycles $\{-1\}$, $\{-5, -7\}$, $\{-17, \dots, -91\}$ have mean halving exponents $1, \frac{3}{2}, \frac{11}{7}$, all $< \alpha$: *the divergence-capable statistics are realized by actual integers* — negative ones. Any proof of non-divergence for positive integers must therefore use the tail sign; no purely distributional argument can suffice. [PROVED]

5 Statistical camouflage

Theorem 5.1 (Symbol independence). [PROVED + VERIFIED] *Under Haar measure on $\mathbb{Z}_2^\times$, the macro-symbol pair (l, K') is exactly independent with $l, K' \sim \text{Geom}(1/2)$; consequently the entire macro-symbol process $(K_t, l_t)_t$ is i.i.d. product-geometric.*

Proof. Let $x = 3^K a$, Haar-uniform odd. If $x \equiv 3 \pmod{4}$ then $l = 1$ and $K' = v_2(x+1) - 1$ is geometric from the higher bits; if $x \equiv 1 \pmod{4}$ then $l = v_2(x-1) \geq 2$ is geometric and $K' = v_2(u+1)$ draws on fresh bits of u . Both branches give independent geometrics; $K_{t+1} = K'_t$ closes the process. (500,000-sample check: all 25 ratios $\Pr(l, K')/[\Pr(l) \Pr(K')] = 1.00 \pm 0.02$.) $\square$

Remark 5.2. All arithmetic of §?? (LTE laws, dead stops, selection rules) lives in the deterministic residue thread and is *invisible* in the symbols: a hidden-variable decomposition of an exactly-product law. This explains why symbol-level or entropy arguments invariably rediscover the same gap constant and cannot by themselves decide the conjecture. Re-entry structure is likewise pure luck: along random orbits with $n \geq 2^{16}$ (96k steps), the shadow-quality tail is exactly geometric, $P(Q \geq q) \propto 2^{-q}$ over nine octaves (successive ratios 0.49–0.52). [MEASURED]

6 Composition of record climbs and the n^2 envelope

Lagarias and Weiss [?] introduced the repeated random walk model and proved, within it, that the maximum excursion satisfies $\limsup \log t(n)/\log n = 2$. We add the explicit tilt. Let $I(m) = D_{\text{KL}}(\text{Geom}(1/m) \parallel \text{Geom}(1/2))$ in bits, the large-deviation cost of tilting the halving mean to m .

Theorem 6.1 (Climb-cost identity). [PROVED + VERIFIED] $I'(m) = \log_2 \frac{2(m-1)}{m}$, and

$$I(\frac{4}{3}) = \log_2 3 - \frac{4}{3}, \quad \text{hence} \quad \min_{1 < m < \alpha} \frac{I(m)}{\alpha - m} = 1, \quad \text{attained at } m^* = \frac{4}{3}.$$

Proof. Differentiate $I(m) = \log_2 \frac{2}{m} + (m-1) \log_2 \frac{2(m-1)}{m}$; the cross terms cancel, leaving $I'(m) = \log_2 \frac{2(m-1)}{m}$. At $m = 4/3$ this is -1 , making $h(m) = I(m) - \alpha + m$ critical there; $h(4/3) = 0$ by direct computation, and convexity of I gives the global minimum. $\square$

Corollary 6.2. [PROVED] (i) *The cheapest sustained climb has halving mean 4/3: by Gibbs conditioning on the exact i.i.d. symbols (Theorem ??), the empirical e-law of a conditioned optimal climb converges to $\text{Geom}(3/4)$ — in particular $e=1$ on 3/4 of climb steps.* (ii) *The cost of E bits of climb is exactly E bits of measure: $P(\text{excursion} \geq E) \asymp 2^{-E}$, whence record peaks scale as n^2 (the Lagarias–Weiss law) with Gumbel correction $E^*(K) = K - \Theta(\log K)$ at seed size K .*

Measurements. [MEASURED] Pooling the climbs of all excursion records with $E \geq 8$ below 2^{23} (728 steps): $\Pr(e=1) = 0.7775$ vs 0.75; $\Pr(e=2) = 0.1854$ vs 0.1875; mean $e = 1.273$ vs $4/3$, with the two *longest* climbs (49 and 51 steps) at 0.714 and 0.725 — short bursts run hot, long climbs sit on the tilt, as the conditional limit demands. The record envelope below 2^{23} fits $E^*(K) = 0.971K - 1.51 \log_2 K + 2.7$; at the largest published scales the ratios $\rho(n)$ are 2.004–2.099 for $n \sim 10^9$ – 10^{18} [?, ?]. Notably the records draw ~ 75 – 80% of their halving budget from *beyond* the free bit window of the seed, yet still meet the Haar envelope: the deterministic zero-tail continuation is statistically typical even at the extremes.

7 Aperiodicity, and what remains

For completeness: an eventually periodic itinerary forces the orbit onto an integer cycle (classical via the rational-periodicity correspondence [?, ?, ?]; a one-line 2-adic proof: shared shrinking cylinders force $|T^t n - x_t|_2 \rightarrow 0$ toward the periodic point, and every cycle is 2-adically repelling since $|(T^A)'|_2 = 2^B > 1$). Hence divergent orbits have aperiodic itineraries. Combining §?? and §??: every climb *mechanism* is classified and none can sustain divergence; what a divergent orbit would require is an aperiodic, unstructured supply of re-entry luck of linearly growing total depth, against an exactly geometric tail — a measure-zero, Hausdorff-dimension $H(1/\alpha) = 0.9507$ event with, by Theorem ??, no arithmetic mechanism available to produce it.

References

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